

分布式计算实验作业 1

1. 实验要求

自选一张图片, 按照实验指南说明在树莓派集群上基于 MPI 对图片进行边缘提取实验, 记录梯度向量幅度的最小值和最大值, 记录并比较串行算法、单节点 OpenMP 并行算法和 MPI 分布式并行算法的运行时间, 并提交处理后的边缘提取结果图片。试一下: 模拟共享存储方案, 将待处理图片拷贝到每个集群节点上, 修改 `sobel_mpi.cpp` 或 `sobel_mpi_omp.cpp`, 主节点和从节点之间只传递少量的图像分区参数和分区梯度幅度的最大、最小值等信息以及收集处理结果, 而不再分发原始图像分块数据 (图像数据直接从本地加载), 比较修改后和修改前的处理速度, 并分析整个算法的效率瓶颈。

2. 实验过程

编译三个文件:

```
renjianxin@raspi00:~/public_html/temp $ g++ -fopenmp -o sobel_omp sobel_omp.cpp -lfreeimage
renjianxin@raspi00:~/public_html/temp $ ls
all-hosts.list          imgutils.h              sobel.cpp
all-nodes.list          jetson-hosts.list       sobel.h
amdahl_test_mpi.sh      jetson-nodes.list       sobel_mpi
batchresize.cpp          logutils.h              sobel_mpi.cpp
cathedral.jpg            oclutils.h              sobel_mpi_np_t.log
earthmap.jpg             other-nodes.list         sobel_mpi_omp.cpp
gustafson_test_mpi.sh    other-raspi-nodes.list  sobel_ocl.cpp
hello_mpi.cpp            raspi-hosts.list        sobel_omp
IBM_Blue_Gene_P_supercomputer.jpg  raspi-nodes.list       sobel_omp.cpp
IBM_Blue_Gene_P_supercomputer_mpi_out.jpg  sobel.cl
renjianxin@raspi00:~/public_html/temp $

renjianxin@raspi00:~/public_html/temp $ g++ -fopenmp -o sobel sobel.cpp -lfreeimage
renjianxin@raspi00:~/public_html/temp $ ls
all-hosts.list          imgutils.h              sobel.cl
all-nodes.list          jetson-hosts.list       sobel.cpp
amdahl_test_mpi.sh      jetson-nodes.list       sobel.h
batchresize.cpp          logutils.h              sobel_mpi
cathedral.jpg            oclutils.h              sobel_mpi.cpp
earthmap.jpg             other-nodes.list         sobel_mpi_np_t.log
gustafson_test_mpi.sh    other-raspi-nodes.list  sobel_mpi_omp.cpp
hello_mpi.cpp            raspi-hosts.list        sobel_ocl.cpp
IBM_Blue_Gene_P_supercomputer.jpg  raspi-nodes.list       sobel_omp
IBM_Blue_Gene_P_supercomputer_mpi_out.jpg  sobel                  sobel_omp.cpp
renjianxin@raspi00:~/public_html/temp $
```

运行 sobel

```
renjianxin@raspi00:~/public_html/temp $ ./sobel IBM_Blue_Gene_P_supercomputer.jpg
Filtering of input image start ...
the minimum value: 0
the maximum value: 1052.84
The total time for execution is 16.4514s
```

运行 sobel_omp

```
renjianxin@raspi00:~/public_html/temp $ ./sobel_omp IBM_Blue_Gene_P_supercomputer.jpg
Filtering of input image start ...
the minimum value: 0
the maximum value: 1052.84
The total time for execution is 4.15126s
```

运行 sobel_mpi

```
renjianxin@raspi00:~/public_html/temp $ mpirun -np 40 --hostfile raspi-hosts.list ./sobel_mpi IBM_Blue_Gene_P_supercomputer.jpg
Filtering of input image on Process 1 of raspi00 start ...
Filtering of input image on Process 2 of raspi00 start ...
Filtering of input image on Process 3 of raspi00 start ...
Filtering of input image on Process 4 of raspi01 start ...
Filtering of input image on Process 5 of raspi01 start ...
Filtering of input image on Process 6 of raspi01 start ...
Filtering of input image on Process 7 of raspi01 start ...
Filtering of input image on Process 8 of raspi02 start ...
Filtering of input image on Process 9 of raspi02 start ...
Filtering of input image on Process 10 of raspi02 start ...
Filtering of input image on Process 11 of raspi02 start ...
Filtering of input image on Process 12 of raspi03 start ...
Filtering of input image on Process 13 of raspi03 start ...
Filtering of input image on Process 14 of raspi03 start ...
Filtering of input image on Process 15 of raspi03 start ...
Filtering of input image on Process 16 of raspi04 start ...
Filtering of input image on Process 17 of raspi04 start ...
Filtering of input image on Process 18 of raspi04 start ...
Filtering of input image on Process 19 of raspi04 start ...
Filtering of input image on Process 20 of raspi10 start ...
Filtering of input image on Process 21 of raspi10 start ...
Filtering of input image on Process 22 of raspi10 start ...
Filtering of input image on Process 24 of raspi11 start ...
Filtering of input image on Process 23 of raspi11 start ...
Filtering of input image on Process 25 of raspi11 start ...
Filtering of input image on Process 26 of raspi11 start ...
Filtering of input image on Process 27 of raspi11 start ...
Filtering of input image on Process 28 of raspi12 start ...
Filtering of input image on Process 29 of raspi12 start ...
Filtering of input image on Process 30 of raspi12 start ...
Filtering of input image on Process 31 of raspi12 start ...
Filtering of input image on Process 32 of raspi13 start ...
Filtering of input image on Process 33 of raspi13 start ...
Filtering of input image on Process 34 of raspi13 start ...
Filtering of input image on Process 0 of raspi00 start ...
Filtering of input image on Process 35 of raspi13 start ...
Filtering of input image on Process 36 of raspi14 start ...
Min/Max on Process 1 of raspi00: 0/845.693
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Filtering of input image on Process 37 of raspi14 start ...
Filtering of input image on Process 38 of raspi14 start ...
Filtering of input image on Process 39 of raspi14 start ...
Min/Max on Process 5 of raspi01: 0/900.681
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 4 of raspi01: 0/760.138
```

```
Receive global min & max from MASTER ...
Min/Max on Process 4 of raspi01: 0/760.138
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 3 of raspi00: 0/840.335
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 6 of raspi01: 0/895.433
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 7 of raspi01: 0/891.977
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 21 of raspi10: 0/1052.84
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 20 of raspi10: 0/952.638
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 22 of raspi10: 0/1021.81
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 2 of raspi00: 0/851.564
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 23 of raspi10: 0/935.916
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 24 of raspi11: 0/935.192
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 25 of raspi11: 0/939.699
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 26 of raspi11: 0/989.947
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 27 of raspi11: 0/963.534
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 0 of raspi00: 0/867.211
Receive min & max from other nodes ...
Min/Max on Process 8 of raspi02: 0/956.114
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 10 of raspi02: 0/1029.34
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 11 of raspi02: 0/831.269
Send min & max to MASTER ...
```

```
Receive global min & max from MASTER ...
Min/Max on Process 11 of raspi02: 0/831.269
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 12 of raspi03: 0/997.255
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 9 of raspi02: 0/1047.14
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 13 of raspi03: 0/1017.99
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 14 of raspi03: 0/1009.54
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 16 of raspi04: 0/1001.96
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 17 of raspi04: 0/1014.87
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 15 of raspi03: 0/934.783
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 18 of raspi04: 0/979.834
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 19 of raspi04: 0/935.331
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 28 of raspi12: 0/1034.06
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 30 of raspi12: 0/1017.39
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 31 of raspi12: 0/1023.91
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 32 of raspi13: 0/1006.28
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 34 of raspi13: 0/895.087
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 29 of raspi12: 0/1014.17
Send min & max to MASTER ...
Receive global min & max from MASTER ...
```



```
Min/Max on Process 33 of raspi13: 0/991.541
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 35 of raspi13: 0/901.077
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 37 of raspi14: 0/526.551
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 39 of raspi14: 0/422.756
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 38 of raspi14: 0/403.777
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 36 of raspi14: 0/758.563
Send min & max to MASTER ...
the global minimum value: 0
the global maximum value: 1052.84
Send global min & max to other nodes ...
Receive global min & max from MASTER ...
Filtering of input image on Process 0 of raspi00 finish.
Filtering of input image on Process 5 of raspi01 finish.
Filtering of input image on Process 26 of raspi11 finish.
Filtering of input image on Process 7 of raspi01 finish.
Filtering of input image on Process 21 of raspi10 finish.
Filtering of input image on Process 22 of raspi10 finish.
Filtering of input image on Process 25 of raspi11 finish.
Filtering of input image on Process 6 of raspi01 finish.
Filtering of input image on Process 27 of raspi11 finish.
Filtering of input image on Process 4 of raspi01 finish.
Filtering of input image on Process 1 of raspi00 finish.
Filtering of input image on Process 24 of raspi11 finish.
Filtering of input image on Process 23 of raspi10 finish.
Filtering of input image on Process 20 of raspi10 finish.
Filtering of input image on Process 3 of raspi00 finish.
Filtering of input image on Process 2 of raspi00 finish.
Filtering of input image on Process 37 of raspi14 finish.
Filtering of input image on Process 31 of raspi12 finish.
Filtering of input image on Process 34 of raspi13 finish.
Filtering of input image on Process 39 of raspi14 finish.
Filtering of input image on Process 32 of raspi13 finish.
Filtering of input image on Process 38 of raspi14 finish.
Filtering of input image on Process 30 of raspi12 finish.
Filtering of input image on Process 36 of raspi14 finish.
Filtering of input image on Process 33 of raspi13 finish.
Filtering of input image on Process 28 of raspi12 finish.
Filtering of input image on Process 8 of raspi02 finish.
Filtering of input image on Process 16 of raspi04 finish.
```

```
Filtering of input image on Process 12 of raspi03 finish.
Filtering of input image on Process 13 of raspi03 finish.
Filtering of input image on Process 11 of raspi02 finish.
Filtering of input image on Process 10 of raspi02 finish.
Filtering of input image on Process 14 of raspi03 finish.
Filtering of input image on Process 17 of raspi04 finish.
Filtering of input image on Process 35 of raspi13 finish.
Filtering of input image on Process 29 of raspi12 finish.
Filtering of input image on Process 19 of raspi04 finish.
Filtering of input image on Process 18 of raspi04 finish.
Filtering of input image on Process 15 of raspi03 finish.
Filtering of input image on Process 9 of raspi02 finish.
The total time for execution is 1.71649s
```

3. 处理结果

Index of / - ×

中国科学院 ×

中国科学院 ×

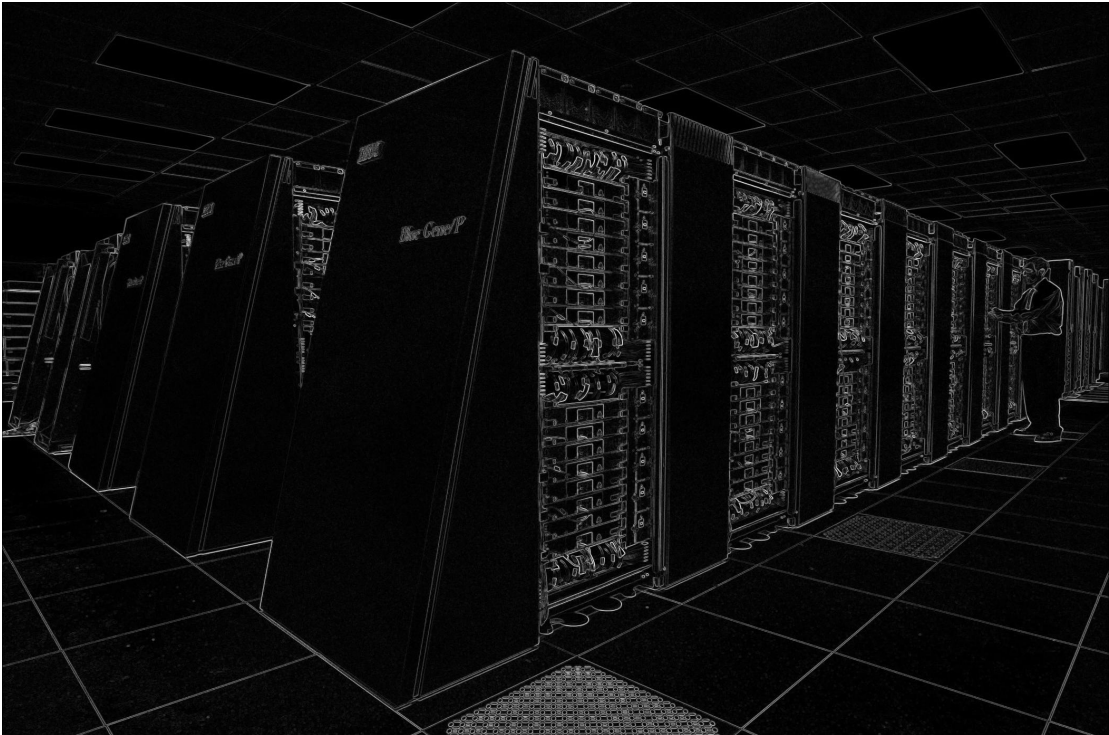
云计算 -

← → ↻ 🏠 ⓘ localhost:36899/~renjianxin/temp/ ⌂

Index of /~renjianxin/temp

	Name	Last modified	Size	Description
📁	Parent Directory		-	
🖼️	IBM Blue Gene P supercomputer.jpg	2026-04-26 17:24	1.7M	
🖼️	IBM Blue Gene P supercomputer mpi_out.jpg	2026-04-26 18:40	405K	
🖼️	IBM Blue Gene P supercomputer omp_out.jpg	2026-04-26 18:40	396K	
🖼️	IBM Blue Gene P supercomputer_out.jpg	2026-04-26 18:40	396K	
📄	all-hosts.list	2026-04-26 17:24	168	
📄	all-nodes.list	2026-04-26 17:24	98	
📄	amdahl_test_mpi.sh	2026-04-26 17:24	139	
📄	...	2026-04-26 17:24	0.0K	

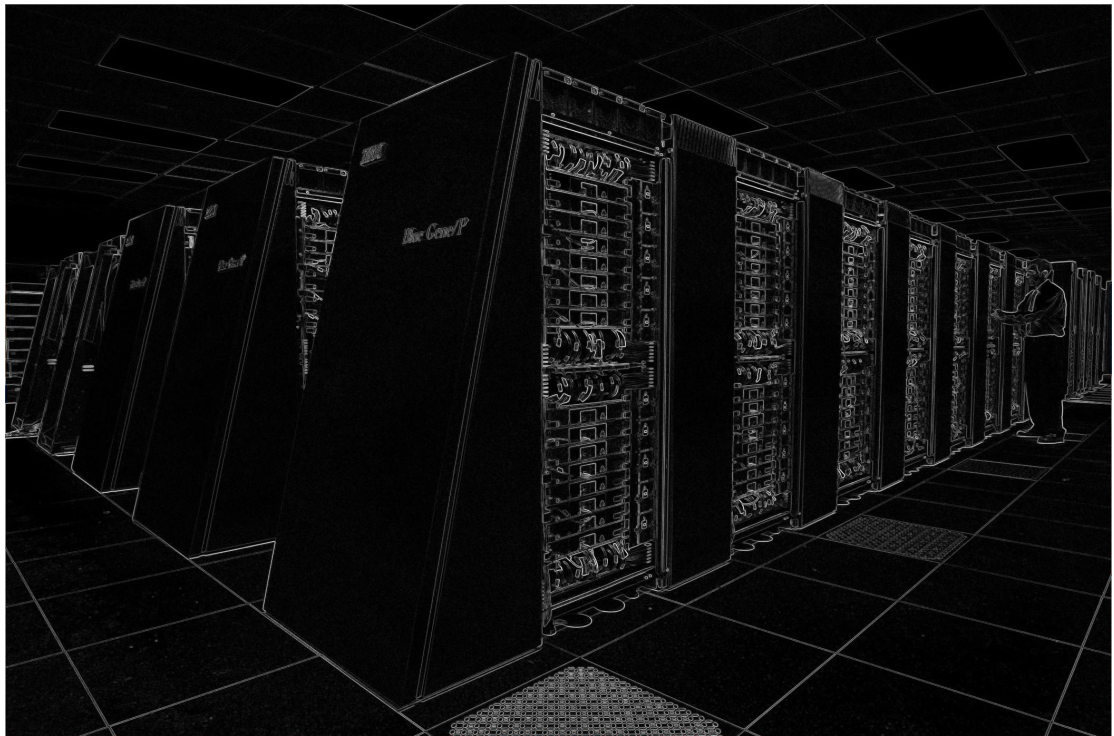
sobel



SOBEL_OMP



SOBEL_MPI



4. 串行算法、单节点 OpenMP 并行算法和 MPI 分布式并行算法比较

算法模式	进程数	运行时间	最小值	最大值
串行	1	16.45s	0	1052.84
OpenMP	4	4.15s	0	1052.84
MPI	40	1.72s	0	1052.84

OpenMp 跟串行相比速度提升了约四倍，MPI 跟串行相比速度提升了约 10 倍。证明了分布式计算处理大数据的优势。

5. 修改 sobel_mpi.cpp 代码后进行编译

```
renjianxin@raspi00:~/public_html/temp $  
renjianxin@raspi00:~/public_html/temp $ mpic++ sobel_test.cpp -o sobel_mpi_new -lfreeimage  
renjianxin@raspi00:~/public_html/temp $ mpirun -np 40 --hostfile raspi-hosts.list ./sobel_mpi_new renjianxin.jpg  
Filtering of input image on Process 2 of raspi00 start ...
```

6. 运行 sobel_mpi

```
renjianxin@raspi00:~/public_html/temp $ mpirun -np 40 --hostfile raspi-hosts.list ./sobel_mpi renjianxin.jpg  
Filtering of input image on Process 1 of raspi00 start ...  
Filtering of input image on Process 2 of raspi00 start ...  
Filtering of input image on Process 3 of raspi00 start ...  
Filtering of input image on Process 4 of raspi01 start ...  
Filtering of input image on Process 5 of raspi01 start ...  
Filtering of input image on Process 6 of raspi01 start ...  
Filtering of input image on Process 7 of raspi01 start ...  
Filtering of input image on Process 8 of raspi02 start ...  
Filtering of input image on Process 9 of raspi02 start ...  
Filtering of input image on Process 10 of raspi02 start ...  
Filtering of input image on Process 11 of raspi02 start ...  
Filtering of input image on Process 12 of raspi03 start ...  
Filtering of input image on Process 13 of raspi03 start ...  
Filtering of input image on Process 14 of raspi03 start ...  
Min/Max on Process 3 of raspi00: 0/443.644  
Send min & max to MASTER ...  
Receive global min & max from MASTER ...  
Filtering of input image on Process 15 of raspi03 start ...  
Min/Max on Process 2 of raspi00: 0/389.597  
Send min & max to MASTER ...  
Receive global min & max from MASTER ...  
Filtering of input image on Process 16 of raspi04 start ...  
Min/Max on Process 5 of raspi01: 0/414.462  
Send min & max to MASTER ...  
Min/Max on Process 4 of raspi01: 0/450.391  
Send min & max to MASTER ...  
Receive global min & max from MASTER ...  
Receive global min & max from MASTER ...  
Min/Max on Process 1 of raspi00: 0/427.13  
Send min & max to MASTER ...  
Receive global min & max from MASTER ...  
Min/Max on Process 6 of raspi01: 0/403.205  
Send min & max to MASTER ...  
Receive global min & max from MASTER ...  
Filtering of input image on Process 17 of raspi04 start ...  
Min/Max on Process 7 of raspi01: 0/362.322  
Send min & max to MASTER ...  
Filtering of input image on Process 18 of raspi04 start ...  
Receive global min & max from MASTER ...  
Filtering of input image on Process 19 of raspi04 start ...  
Filtering of input image on Process 20 of raspi10 start ...  
Filtering of input image on Process 21 of raspi10 start ...  
Filtering of input image on Process 22 of raspi10 start ...  
Filtering of input image on Process 23 of raspi10 start ...  
Filtering of input image on Process 24 of raspi11 start ...  
Filtering of input image on Process 25 of raspi11 start ...  
Filtering of input image on Process 26 of raspi11 start ...
```

```
Filtering of input image on Process 26 of raspi11 start ...
Filtering of input image on Process 27 of raspi11 start ...
Filtering of input image on Process 28 of raspi12 start ...
Filtering of input image on Process 29 of raspi12 start ...
Filtering of input image on Process 30 of raspi12 start ...
Filtering of input image on Process 31 of raspi12 start ...
Filtering of input image on Process 32 of raspi13 start ...
Min/Max on Process 20 of raspi0: 0/335.339
Send min & max to MASTER ...
Min/Max on Process 21 of raspi0: 0/371.892
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Receive global min & max from MASTER ...
Filtering of input image on Process 33 of raspi13 start ...
Min/Max on Process 8 of raspi02: 0/421.882
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 22 of raspi0: 0/302.069
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Filtering of input image on Process 34 of raspi13 start ...
Min/Max on Process 23 of raspi0: 0/293.782
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 24 of raspi11: 0/300.585
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Filtering of input image on Process 35 of raspi13 start ...
Min/Max on Process 10 of raspi02: 0/364.335
Send min & max to MASTER ...
Filtering of input image on Process 36 of raspi14 start ...
Receive global min & max from MASTER ...
Min/Max on Process 25 of raspi11: 0/341.836
Send min & max to MASTER ...
Filtering of input image on Process 0 of raspi00 start ...
Receive global min & max from MASTER ...
Min/Max on Process 9 of raspi02: 0/366.463
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 26 of raspi11: 0/249.87
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Filtering of input image on Process 37 of raspi14 start ...
Min/Max on Process 27 of raspi11: 0/282.385
Send min & max to MASTER ...
Filtering of input image on Process 38 of raspi14 start ...
Receive global min & max from MASTER ...
Filtering of input image on Process 39 of raspi14 start ...
Min/Max on Process 11 of raspi02: 0/361.124
```

```
Receive global min & max from MASTER ...
Filtering of input image on Process 39 of raspi14 start ...
Min/Max on Process 11 of raspi02: 0/361.124
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 13 of raspi03: 0/385.423
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 12 of raspi03: 0/403.091
Send min & max to MASTER ...
Min/Max on Process 14 of raspi03: 0/407.355
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 16 of raspi04: 0/446
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 15 of raspi03: 0/449.308
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 18 of raspi04: 0/429.677
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 17 of raspi04: 0/454.348
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 0 of raspi00: 0/482.004
Receive min & max from other nodes ...
Min/Max on Process 19 of raspi04: 0/429.631
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 28 of raspi12: 0/331.255
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 29 of raspi12: 0/269.706
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 30 of raspi12: 0/262.467
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 32 of raspi13: 0/252.153
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 33 of raspi13: 0/246.075
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 31 of raspi12: 0/263.743
Send min & max to MASTER ...
Receive global min & max from MASTER ...
```

```
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 34 of raspi13: 0/271.78
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 36 of raspi14: 0/316.017
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 35 of raspi13: 0/294.957
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 38 of raspi14: 0/332.936
Send min & max to MASTER ...
Min/Max on Process 37 of raspi14: 0/259.293
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 39 of raspi14: 0/306.161
Send min & max to MASTER ...
the global minimum value: 0
the global maximum value: 482.004
Send global min & max to other nodes ...
Receive global min & max from MASTER ...
Filtering of input image on Process 0 of raspi00 finish.
Filtering of input image on Process 3 of raspi00 finish.
Filtering of input image on Process 2 of raspi00 finish.
Filtering of input image on Process 5 of raspi01 finish.
Filtering of input image on Process 25 of raspi11 finish.
Filtering of input image on Process 24 of raspi11 finish.
Filtering of input image on Process 22 of raspi10 finish.
Filtering of input image on Process 1 of raspi00 finish.
Filtering of input image on Process 23 of raspi10 finish.
Filtering of input image on Process 7 of raspi01 finish.
Filtering of input image on Process 21 of raspi10 finish.
Filtering of input image on Process 6 of raspi01 finish.
Filtering of input image on Process 4 of raspi01 finish.
Filtering of input image on Process 26 of raspi11 finish.
Filtering of input image on Process 27 of raspi11 finish.
Filtering of input image on Process 20 of raspi10 finish.
Filtering of input image on Process 39 of raspi14 finish.
Filtering of input image on Process 28 of raspi12 finish.
Filtering of input image on Process 8 of raspi02 finish.
Filtering of input image on Process 11 of raspi02 finish.
Filtering of input image on Process 32 of raspi13 finish.
Filtering of input image on Process 33 of raspi13 finish.
Filtering of input image on Process 16 of raspi04 finish.
Filtering of input image on Process 14 of raspi03 finish.
Filtering of input image on Process 13 of raspi03 finish.

Filtering of input image on Process 12 of raspi03 finish.
Filtering of input image on Process 17 of raspi04 finish.
Filtering of input image on Process 10 of raspi02 finish.
Filtering of input image on Process 9 of raspi02 finish.
Filtering of input image on Process 15 of raspi03 finish.
Filtering of input image on Process 18 of raspi04 finish.
Filtering of input image on Process 19 of raspi04 finish.
Filtering of input image on Process 35 of raspi13 finish.
Filtering of input image on Process 34 of raspi13 finish.
Filtering of input image on Process 30 of raspi12 finish.
Filtering of input image on Process 29 of raspi12 finish.
Filtering of input image on Process 31 of raspi12 finish.
Filtering of input image on Process 36 of raspi14 finish.
Filtering of input image on Process 38 of raspi14 finish.
Filtering of input image on Process 37 of raspi14 finish.
The total time for execution is 0.545944s
```

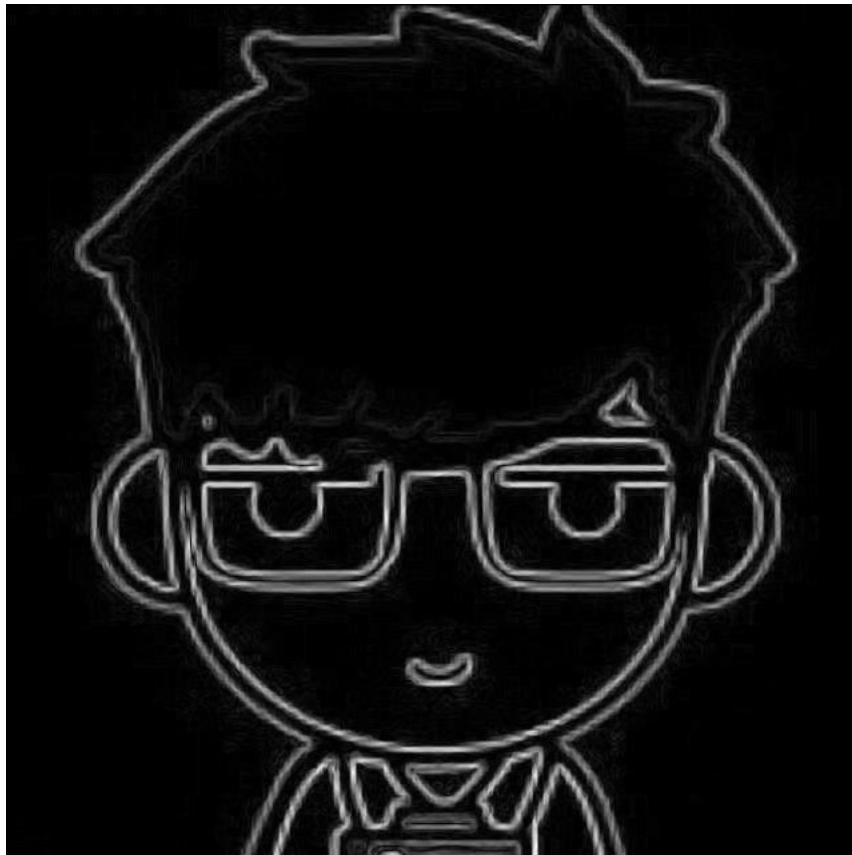



7. 运行 sobel_mpi_new

```
renjianxin@raspi00:~/public_html/temp $ mpirun -np 40 --hostfile raspi-hosts.list ./sobel_mpi_new renjianxin.jpg
Filtering of input image on Process 2 of raspi00 start ...
Filtering of input image on Process 8 of raspi02 start ...
Filtering of input image on Process 6 of raspi01 start ...
Filtering of input image on Process 10 of raspi02 start ...
Filtering of input image on Process 24 of raspi11 start ...
Filtering of input image on Process 22 of raspi10 start ...
Filtering of input image on Process 26 of raspi11 start ...
Filtering of input image on Process 4 of raspi01 start ...
Filtering of input image on Process 20 of raspi10 start ...
Filtering of input image on Process 16 of raspi04 start ...
Filtering of input image on Process 32 of raspi13 start ...
Filtering of input image on Process 38 of raspi14 start ...
Filtering of input image on Process 18 of raspi04 start ...
Filtering of input image on Process 34 of raspi13 start ...
Filtering of input image on Process 36 of raspi14 start ...
Filtering of input image on Process 5 of raspi01 start ...
Filtering of input image on Process 9 of raspi02 start ...
Filtering of input image on Process 21 of raspi10 start ...
Filtering of input image on Process 3 of raspi00 start ...
Filtering of input image on Process 25 of raspi11 start ...
Filtering of input image on Process 1 of raspi00 start ...
Filtering of input image on Process 11 of raspi02 start ...
Filtering of input image on Process 17 of raspi04 start ...
Filtering of input image on Process 27 of raspi11 start ...
Filtering of input image on Process 7 of raspi01 start ...
Filtering of input image on Process 37 of raspi14 start ...
Filtering of input image on Process 23 of raspi10 start ...
Filtering of input image on Process 33 of raspi13 start ...
Filtering of input image on Process 35 of raspi13 start ...
Filtering of input image on Process 19 of raspi04 start ...
Filtering of input image on Process 39 of raspi14 start ...
Filtering of input image on Process 0 of raspi00 start ...
Filtering of input image on Process 13 of raspi03 start ...
Filtering of input image on Process 14 of raspi03 start ...
Filtering of input image on Process 29 of raspi12 start ...
Filtering of input image on Process 30 of raspi12 start ...
Filtering of input image on Process 15 of raspi03 start ...
Filtering of input image on Process 31 of raspi12 start ...
Filtering of input image on Process 12 of raspi03 start ...
Filtering of input image on Process 28 of raspi12 start ...
Local Min/Max on Process 8: 0/482.004
Local Min/Max on Process 6: 0/482.004
Local Min/Max on Process 22: 0/482.004
Local Min/Max on Process 24: 0/482.004
Local Min/Max on Process 25: 0/482.004
Local Min/Max on Process 26: 0/482.004
Local Min/Max on Process 9: 0/482.004
Local Min/Max on Process 2: 0/482.004
```

```
Local Min/Max on Process 8: 0/482.004
Local Min/Max on Process 6: 0/482.004
Local Min/Max on Process 22: 0/482.004
Local Min/Max on Process 24: 0/482.004
Local Min/Max on Process 25: 0/482.004
Local Min/Max on Process 26: 0/482.004
Local Min/Max on Process 9: 0/482.004
Local Min/Max on Process 2: 0/482.004
Local Min/Max on Process 23: 0/482.004
Local Min/Max on Process 7: 0/482.004
Local Min/Max on Process 10: 0/482.004
Local Min/Max on Process 5: 0/482.004
Local Min/Max on Process 21: 0/482.004
Local Min/Max on Process 3: 0/482.004
Local Min/Max on Process 4: 0/482.004
Local Min/Max on Process 11: 0/482.004
Local Min/Max on Process 20: 0/482.004
Local Min/Max on Process 27: 0/482.004
Local Min/Max on Process 0: 0/482.004
Local Min/Max on Process 1: 0/482.004
Local Min/Max on Process 36: 0/482.004
Local Min/Max on Process 32: 0/482.004
Local Min/Max on Process 17: 0/482.004
Local Min/Max on Process 18: 0/482.004
Local Min/Max on Process 14: 0/482.004
Local Min/Max on Process 31: 0/482.004
Local Min/Max on Process 15: 0/482.004
Local Min/Max on Process 29: 0/482.004
Local Min/Max on Process 35: 0/482.004
Local Min/Max on Process 34: 0/482.004
Local Min/Max on Process 16: 0/482.004
Local Min/Max on Process 37: 0/482.004
Local Min/Max on Process 38: 0/482.004
Local Min/Max on Process 13: 0/482.004
Local Min/Max on Process 30: 0/482.004
Local Min/Max on Process 33: 0/482.004
Local Min/Max on Process 28: 0/482.004
Local Min/Max on Process 39: 0/482.004
Local Min/Max on Process 12: 0/482.004
Local Min/Max on Process 19: 0/482.004
Global Min/Max: 0/482.004
Filtering of input image on Process 6 finish.
Filtering of input image on Process 8 finish.
```

```
Global Min/Max: 0/482.004
Filtering of input image on Process 6 finish.
Filtering of input image on Process 8 finish.
Filtering of input image on Process 23 finish.
Filtering of input image on Process 3 finish.
Filtering of input image on Process 9 finish.
Filtering of input image on Process 25 finish.
Filtering of input image on Process 7 finish.
Filtering of input image on Process 22 finish.
Filtering of input image on Process 2 finish.
Filtering of input image on Process 24 finish.
Filtering of input image on Process 5 finish.
Filtering of input image on Process 21 finish.
Filtering of input image on Process 26 finish.
Filtering of input image on Process 11 finish.
Filtering of input image on Process 4 finish.
Filtering of input image on Process 20 finish.
Filtering of input image on Process 27 finish.
Filtering of input image on Process 10 finish.
Filtering of input image on Process 1 finish.
Filtering of input image on Process 0 finish.
The total time for execution is 2.3998s
Saved output to renjianxin_local_mpi_out.jpg
Filtering of input image on Process 29 finish.
Filtering of input image on Process 31 finish.
Filtering of input image on Process 30 finish.
Filtering of input image on Process 28 finish.
Filtering of input image on Process 36 finish.
Filtering of input image on Process 37 finish.
Filtering of input image on Process 38 finish.
Filtering of input image on Process 15 finish.
Filtering of input image on Process 39 finish.
Filtering of input image on Process 14 finish.
Filtering of input image on Process 13 finish.
Filtering of input image on Process 12 finish.
Filtering of input image on Process 32 finish.
Filtering of input image on Process 35 finish.
Filtering of input image on Process 34 finish.
Filtering of input image on Process 33 finish.
Filtering of input image on Process 17 finish.
Filtering of input image on Process 18 finish.
Filtering of input image on Process 16 finish.
Filtering of input image on Process 19 finish.
```



8. 对比分析

测试图片	方 案 A (sobel_mpi_new) (模拟共享存储/本地 加载)	方案 B (sobel_mpi) (主节点分发数据)	结果对比 (A / B)
renjianxin.jpg	2.40 s	0.54 s	A 慢了 4.4 倍
cathedral.jpg	崩溃	22.77 s	A 无法运行

在当前的树莓派的集群下，网络传输的效率远高于所有节点同时读取本地存储的效率。
问题可能出现在存储瓶颈、内存瓶颈。sobel_mpi_new 尝试 40 个进程同时处理 cathedral.jpg 时结果崩溃。最终出现了反直觉的现象。

9. log 信息

```
renjianxin@raspi00:~/public_html/temp $ mpirun -np 40 --hostfile
raspi-hosts.list ./sobel_mpi_new renjianxin.jpg
Filtering of input image on Process 2 of raspi00 start ...
Filtering of input image on Process 8 of raspi02 start ...
Filtering of input image on Process 6 of raspi01 start ...
Filtering of input image on Process 10 of raspi02 start ...
Filtering of input image on Process 24 of raspi11 start ...
Filtering of input image on Process 22 of raspi10 start ...
Filtering of input image on Process 26 of raspi11 start ...
Filtering of input image on Process 4 of raspi01 start ...
Filtering of input image on Process 20 of raspi10 start ...
Filtering of input image on Process 16 of raspi04 start ...
Filtering of input image on Process 32 of raspi13 start ...
Filtering of input image on Process 38 of raspi14 start ...
Filtering of input image on Process 18 of raspi04 start ...
Filtering of input image on Process 34 of raspi13 start ...
Filtering of input image on Process 36 of raspi14 start ...
Filtering of input image on Process 5 of raspi01 start ...
Filtering of input image on Process 9 of raspi02 start ...
Filtering of input image on Process 21 of raspi10 start ...
Filtering of input image on Process 3 of raspi00 start ...
Filtering of input image on Process 25 of raspi11 start ...
Filtering of input image on Process 1 of raspi00 start ...
Filtering of input image on Process 11 of raspi02 start ...
Filtering of input image on Process 17 of raspi04 start ...
```

Filtering of input image on Process 27 of raspi11 start ...
Filtering of input image on Process 7 of raspi01 start ...
Filtering of input image on Process 37 of raspi14 start ...
Filtering of input image on Process 23 of raspi10 start ...
Filtering of input image on Process 33 of raspi13 start ...
Filtering of input image on Process 35 of raspi13 start ...
Filtering of input image on Process 19 of raspi04 start ...
Filtering of input image on Process 39 of raspi14 start ...
Filtering of input image on Process 0 of raspi00 start ...
Filtering of input image on Process 13 of raspi03 start ...
Filtering of input image on Process 14 of raspi03 start ...
Filtering of input image on Process 29 of raspi12 start ...
Filtering of input image on Process 30 of raspi12 start ...
Filtering of input image on Process 15 of raspi03 start ...
Filtering of input image on Process 31 of raspi12 start ...
Filtering of input image on Process 12 of raspi03 start ...
Filtering of input image on Process 28 of raspi12 start ...
Local Min/Max on Process 8: 0/482.004
Local Min/Max on Process 6: 0/482.004
Local Min/Max on Process 22: 0/482.004
Local Min/Max on Process 24: 0/482.004
Local Min/Max on Process 25: 0/482.004
Local Min/Max on Process 26: 0/482.004
Local Min/Max on Process 9: 0/482.004
Local Min/Max on Process 2: 0/482.004
Local Min/Max on Process 23: 0/482.004
Local Min/Max on Process 7: 0/482.004
Local Min/Max on Process 10: 0/482.004
Local Min/Max on Process 5: 0/482.004
Local Min/Max on Process 21: 0/482.004
Local Min/Max on Process 3: 0/482.004
Local Min/Max on Process 4: 0/482.004
Local Min/Max on Process 11: 0/482.004
Local Min/Max on Process 20: 0/482.004
Local Min/Max on Process 27: 0/482.004
Local Min/Max on Process 0: 0/482.004
Local Min/Max on Process 1: 0/482.004
Local Min/Max on Process 36: 0/482.004
Local Min/Max on Process 32: 0/482.004
Local Min/Max on Process 17: 0/482.004
Local Min/Max on Process 18: 0/482.004
Local Min/Max on Process 14: 0/482.004
Local Min/Max on Process 31: 0/482.004
Local Min/Max on Process 15: 0/482.004

Local Min/Max on Process 29: 0/482.004
Local Min/Max on Process 35: 0/482.004
Local Min/Max on Process 34: 0/482.004
Local Min/Max on Process 16: 0/482.004
Local Min/Max on Process 37: 0/482.004
Local Min/Max on Process 38: 0/482.004
Local Min/Max on Process 13: 0/482.004
Local Min/Max on Process 30: 0/482.004
Local Min/Max on Process 33: 0/482.004
Local Min/Max on Process 28: 0/482.004
Local Min/Max on Process 39: 0/482.004
Local Min/Max on Process 12: 0/482.004
Local Min/Max on Process 19: 0/482.004
Global Min/Max: 0/482.004
Filtering of input image on Process 6 finish.
Filtering of input image on Process 8 finish.
Filtering of input image on Process 23 finish.
Filtering of input image on Process 3 finish.
Filtering of input image on Process 9 finish.
Filtering of input image on Process 25 finish.
Filtering of input image on Process 7 finish.
Filtering of input image on Process 22 finish.
Filtering of input image on Process 2 finish.
Filtering of input image on Process 24 finish.
Filtering of input image on Process 5 finish.
Filtering of input image on Process 21 finish.
Filtering of input image on Process 26 finish.
Filtering of input image on Process 11 finish.
Filtering of input image on Process 4 finish.
Filtering of input image on Process 20 finish.
Filtering of input image on Process 27 finish.
Filtering of input image on Process 10 finish.
Filtering of input image on Process 1 finish.
Filtering of input image on Process 0 finish.
The total time for execution is 2.3998s
Saved output to renjianxin_local_mpi_out.jpg
Filtering of input image on Process 29 finish.
Filtering of input image on Process 31 finish.
Filtering of input image on Process 30 finish.
Filtering of input image on Process 28 finish.
Filtering of input image on Process 36 finish.
Filtering of input image on Process 37 finish.
Filtering of input image on Process 38 finish.
Filtering of input image on Process 15 finish.

Filtering of input image on Process 39 finish.
Filtering of input image on Process 14 finish.
Filtering of input image on Process 13 finish.
Filtering of input image on Process 12 finish.
Filtering of input image on Process 32 finish.
Filtering of input image on Process 35 finish.
Filtering of input image on Process 34 finish.
Filtering of input image on Process 33 finish.
Filtering of input image on Process 17 finish.
Filtering of input image on Process 18 finish.
Filtering of input image on Process 16 finish.
Filtering of input image on Process 19 finish.
renjianxin@raspi00:~/public_html/temp \$ mpirun -np 40 --hostfile
raspi-hosts.list ./sobel_mpi renjianxin.jpg
Filtering of input image on Process 1 of raspi00 start ...
Filtering of input image on Process 2 of raspi00 start ...
Filtering of input image on Process 3 of raspi00 start ...
Filtering of input image on Process 4 of raspi01 start ...
Filtering of input image on Process 5 of raspi01 start ...
Filtering of input image on Process 6 of raspi01 start ...
Filtering of input image on Process 7 of raspi01 start ...
Filtering of input image on Process 8 of raspi02 start ...
Filtering of input image on Process 9 of raspi02 start ...
Filtering of input image on Process 10 of raspi02 start ...
Filtering of input image on Process 11 of raspi02 start ...
Filtering of input image on Process 12 of raspi03 start ...
Filtering of input image on Process 13 of raspi03 start ...
Filtering of input image on Process 14 of raspi03 start ...
Min/Max on Process 3 of raspi00: 0/443.644
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Filtering of input image on Process 15 of raspi03 start ...
Min/Max on Process 2 of raspi00: 0/389.597
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Filtering of input image on Process 16 of raspi04 start ...
Min/Max on Process 5 of raspi01: 0/414.462
Send min & max to MASTER ...
Min/Max on Process 4 of raspi01: 0/450.391
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 1 of raspi00: 0/427.13
Send min & max to MASTER ...

Receive global min & max from MASTER ...
Min/Max on Process 6 of raspi01: 0/403.205
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Filtering of input image on Process 17 of raspi04 start ...
Min/Max on Process 7 of raspi01: 0/362.322
Send min & max to MASTER ...
Filtering of input image on Process 18 of raspi04 start ...
Receive global min & max from MASTER ...
Filtering of input image on Process 19 of raspi04 start ...
Filtering of input image on Process 20 of raspi10 start ...
Filtering of input image on Process 21 of raspi10 start ...
Filtering of input image on Process 22 of raspi10 start ...
Filtering of input image on Process 23 of raspi10 start ...
Filtering of input image on Process 24 of raspi11 start ...
Filtering of input image on Process 25 of raspi11 start ...
Filtering of input image on Process 26 of raspi11 start ...
Filtering of input image on Process 27 of raspi11 start ...
Filtering of input image on Process 28 of raspi12 start ...
Filtering of input image on Process 29 of raspi12 start ...
Filtering of input image on Process 30 of raspi12 start ...
Filtering of input image on Process 31 of raspi12 start ...
Filtering of input image on Process 32 of raspi13 start ...
Min/Max on Process 20 of raspi10: 0/335.339
Send min & max to MASTER ...
Min/Max on Process 21 of raspi10: 0/371.892
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Receive global min & max from MASTER ...
Filtering of input image on Process 33 of raspi13 start ...
Min/Max on Process 8 of raspi02: 0/421.882
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 22 of raspi10: 0/302.069
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Filtering of input image on Process 34 of raspi13 start ...
Min/Max on Process 23 of raspi10: 0/293.782
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 24 of raspi11: 0/300.585
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Filtering of input image on Process 35 of raspi13 start ...

Min/Max on Process 10 of raspi02: 0/364.335
Send min & max to MASTER ...
Filtering of input image on Process 36 of raspi14 start ...
Receive global min & max from MASTER ...
Min/Max on Process 25 of raspi11: 0/341.836
Send min & max to MASTER ...
Filtering of input image on Process 0 of raspi00 start ...
Receive global min & max from MASTER ...
Min/Max on Process 9 of raspi02: 0/366.463
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 26 of raspi11: 0/249.87
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Filtering of input image on Process 37 of raspi14 start ...
Min/Max on Process 27 of raspi11: 0/282.385
Send min & max to MASTER ...
Filtering of input image on Process 38 of raspi14 start ...
Receive global min & max from MASTER ...
Filtering of input image on Process 39 of raspi14 start ...
Min/Max on Process 11 of raspi02: 0/361.124
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 13 of raspi03: 0/385.423
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 12 of raspi03: 0/403.091
Send min & max to MASTER ...
Min/Max on Process 14 of raspi03: 0/407.355
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 16 of raspi04: 0/446
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 15 of raspi03: 0/449.308
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 18 of raspi04: 0/429.677
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 17 of raspi04: 0/454.348
Send min & max to MASTER ...
Receive global min & max from MASTER ...

Min/Max on Process 0 of raspi00: 0/482.004
Receive min & max from other nodes ...
Min/Max on Process 19 of raspi04: 0/429.631
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 28 of raspi12: 0/331.255
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 29 of raspi12: 0/269.706
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 30 of raspi12: 0/262.467
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 32 of raspi13: 0/252.153
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 33 of raspi13: 0/246.075
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 31 of raspi12: 0/263.743
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 34 of raspi13: 0/271.78
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 36 of raspi14: 0/316.017
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 35 of raspi13: 0/294.957
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 38 of raspi14: 0/332.936
Send min & max to MASTER ...
Min/Max on Process 37 of raspi14: 0/259.293
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 39 of raspi14: 0/306.161
Send min & max to MASTER ...
the global minimum value: 0
the global maximum value: 482.004
Send global min & max to other nodes ...
Receive global min & max from MASTER ...

Filtering of input image on Process 0 of raspi00 finish.
Filtering of input image on Process 3 of raspi00 finish.
Filtering of input image on Process 2 of raspi00 finish.
Filtering of input image on Process 5 of raspi01 finish.
Filtering of input image on Process 25 of raspi11 finish.
Filtering of input image on Process 24 of raspi11 finish.
Filtering of input image on Process 22 of raspi10 finish.
Filtering of input image on Process 1 of raspi00 finish.
Filtering of input image on Process 23 of raspi10 finish.
Filtering of input image on Process 7 of raspi01 finish.
Filtering of input image on Process 21 of raspi10 finish.
Filtering of input image on Process 6 of raspi01 finish.
Filtering of input image on Process 4 of raspi01 finish.
Filtering of input image on Process 26 of raspi11 finish.
Filtering of input image on Process 27 of raspi11 finish.
Filtering of input image on Process 20 of raspi10 finish.
Filtering of input image on Process 39 of raspi14 finish.
Filtering of input image on Process 28 of raspi12 finish.
Filtering of input image on Process 8 of raspi02 finish.
Filtering of input image on Process 11 of raspi02 finish.
Filtering of input image on Process 32 of raspi13 finish.
Filtering of input image on Process 33 of raspi13 finish.
Filtering of input image on Process 16 of raspi04 finish.
Filtering of input image on Process 14 of raspi03 finish.
Filtering of input image on Process 13 of raspi03 finish.
Filtering of input image on Process 12 of raspi03 finish.
Filtering of input image on Process 17 of raspi04 finish.
Filtering of input image on Process 10 of raspi02 finish.
Filtering of input image on Process 9 of raspi02 finish.
Filtering of input image on Process 15 of raspi03 finish.
Filtering of input image on Process 18 of raspi04 finish.
Filtering of input image on Process 19 of raspi04 finish.
Filtering of input image on Process 35 of raspi13 finish.
Filtering of input image on Process 34 of raspi13 finish.
Filtering of input image on Process 30 of raspi12 finish.
Filtering of input image on Process 29 of raspi12 finish.
Filtering of input image on Process 31 of raspi12 finish.
Filtering of input image on Process 36 of raspi14 finish.
Filtering of input image on Process 38 of raspi14 finish.
Filtering of input image on Process 37 of raspi14 finish.
The total time for execution is 0.545944s
renjianxin@raspi00:~/public_html/temp \$ mpirun -np 40 --hostfile
raspi-hosts.list ./sobel_mpi cathedral.jpg
Filtering of input image on Process 1 of raspi00 start ...

Filtering of input image on Process 2 of raspi00 start ...
Filtering of input image on Process 3 of raspi00 start ...
Filtering of input image on Process 4 of raspi01 start ...
Filtering of input image on Process 5 of raspi01 start ...
Filtering of input image on Process 6 of raspi01 start ...
Filtering of input image on Process 7 of raspi01 start ...
Filtering of input image on Process 8 of raspi02 start ...
Filtering of input image on Process 9 of raspi02 start ...
Filtering of input image on Process 10 of raspi02 start ...
Filtering of input image on Process 11 of raspi02 start ...
Filtering of input image on Process 12 of raspi03 start ...
Filtering of input image on Process 13 of raspi03 start ...
Filtering of input image on Process 14 of raspi03 start ...
Filtering of input image on Process 15 of raspi03 start ...
Filtering of input image on Process 16 of raspi04 start ...
Filtering of input image on Process 17 of raspi04 start ...
Filtering of input image on Process 18 of raspi04 start ...
Filtering of input image on Process 19 of raspi04 start ...
Filtering of input image on Process 20 of raspi10 start ...
Filtering of input image on Process 21 of raspi10 start ...
Filtering of input image on Process 22 of raspi10 start ...
Filtering of input image on Process 23 of raspi10 start ...
Filtering of input image on Process 24 of raspi11 start ...
Filtering of input image on Process 25 of raspi11 start ...
Filtering of input image on Process 26 of raspi11 start ...
Filtering of input image on Process 27 of raspi11 start ...
Filtering of input image on Process 28 of raspi12 start ...
Filtering of input image on Process 29 of raspi12 start ...
Filtering of input image on Process 30 of raspi12 start ...
Filtering of input image on Process 31 of raspi12 start ...
Filtering of input image on Process 32 of raspi13 start ...
Filtering of input image on Process 33 of raspi13 start ...
Filtering of input image on Process 34 of raspi13 start ...
Filtering of input image on Process 35 of raspi13 start ...
Filtering of input image on Process 36 of raspi14 start ...
Filtering of input image on Process 37 of raspi14 start ...
Filtering of input image on Process 38 of raspi14 start ...
Filtering of input image on Process 0 of raspi00 start ...
Filtering of input image on Process 39 of raspi14 start ...
Min/Max on Process 3 of raspi00: 0/611.471
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 4 of raspi01: 0/592.016
Send min & max to MASTER ...

Receive global min & max from MASTER ...
Min/Max on Process 1 of raspi00: 0/588.555
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 5 of raspi01: 0/583.691
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 6 of raspi01: 0/690.076
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 8 of raspi02: 0/585.109
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 9 of raspi02: 0/650.406
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 7 of raspi01: 0/655.724
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 10 of raspi02: 0/657.755
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 11 of raspi02: 0/637.684
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 20 of raspi10: 0/617.351
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 21 of raspi10: 0/645.871
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 22 of raspi10: 0/611.675
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 23 of raspi10: 0/599.748
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 24 of raspi11: 0/599.04
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 25 of raspi11: 0/575.727
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 26 of raspi11: 0/620.758

Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 27 of raspi11: 0/662.765
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 2 of raspi00: 0/637.377
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 29 of raspi12: 0/578.647
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 28 of raspi12: 0/676.956
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 30 of raspi12: 0/601.169
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 31 of raspi12: 0/605.184
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 32 of raspi13: 0/632.574
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 33 of raspi13: 0/684.814
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 34 of raspi13: 0/607.99
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 35 of raspi13: 0/492.224
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 12 of raspi03: 0/572.694
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 13 of raspi03: 0/545.254
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 14 of raspi03: 0/568.907
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 15 of raspi03: 0/592.014
Send min & max to MASTER ...
Receive global min & max from MASTER ...

Min/Max on Process 0 of raspi00: 0/506.542
Receive min & max from other nodes ...
Min/Max on Process 36 of raspi14: 0/455.338
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 37 of raspi14: 0/507.616
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 38 of raspi14: 0/550.899
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 39 of raspi14: 0/564.668
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 16 of raspi04: 0/622.021
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 17 of raspi04: 0/600.319
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 19 of raspi04: 0/596.658
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 18 of raspi04: 0/610.801
Send min & max to MASTER ...
the global minimum value: 0
the global maximum value: 690.076
Receive global min & max from MASTER ...
Send global min & max to other nodes ...
Filtering of input image on Process 30 of raspi12 finish.
Filtering of input image on Process 35 of raspi13 finish.
Filtering of input image on Process 32 of raspi13 finish.
Filtering of input image on Process 31 of raspi12 finish.
Filtering of input image on Process 29 of raspi12 finish.
Filtering of input image on Process 34 of raspi13 finish.
Filtering of input image on Process 20 of raspi10 finish.
Filtering of input image on Process 22 of raspi10 finish.
Filtering of input image on Process 27 of raspi11 finish.
Filtering of input image on Process 33 of raspi13 finish.
Filtering of input image on Process 24 of raspi11 finish.
Filtering of input image on Process 26 of raspi11 finish.
Filtering of input image on Process 23 of raspi10 finish.
Filtering of input image on Process 28 of raspi12 finish.
Filtering of input image on Process 21 of raspi10 finish.

Filtering of input image on Process 25 of raspi11 finish.
Filtering of input image on Process 14 of raspi03 finish.
Filtering of input image on Process 13 of raspi03 finish.
Filtering of input image on Process 12 of raspi03 finish.
Filtering of input image on Process 15 of raspi03 finish.
Filtering of input image on Process 0 of raspi00 finish.
Filtering of input image on Process 10 of raspi02 finish.
Filtering of input image on Process 6 of raspi01 finish.
Filtering of input image on Process 9 of raspi02 finish.
Filtering of input image on Process 4 of raspi01 finish.
Filtering of input image on Process 5 of raspi01 finish.
Filtering of input image on Process 8 of raspi02 finish.
Filtering of input image on Process 7 of raspi01 finish.
Filtering of input image on Process 11 of raspi02 finish.
Filtering of input image on Process 3 of raspi00 finish.
Filtering of input image on Process 1 of raspi00 finish.
Filtering of input image on Process 2 of raspi00 finish.
Filtering of input image on Process 37 of raspi14 finish.
Filtering of input image on Process 38 of raspi14 finish.
Filtering of input image on Process 36 of raspi14 finish.
Filtering of input image on Process 39 of raspi14 finish.
Filtering of input image on Process 19 of raspi04 finish.
Filtering of input image on Process 17 of raspi04 finish.
Filtering of input image on Process 16 of raspi04 finish.
Filtering of input image on Process 18 of raspi04 finish.

The total time for execution is 22.7715s

```
renjianxin@raspi00:~/public_html/temp $ mpirun -np 40 --hostfile
raspi-hosts.list ./sobel_mpi_new cathedral
cathedral.jpg          cathedral_mpi_out.jpg
renjianxin@raspi00:~/public_html/temp $ mpirun -np 40 --hostfile
raspi-hosts.list ./sobel_mpi_new cathedral.jpg
```

mpirun noticed that process rank 3 with PID 0 on node raspi00 exited on signal 9 (Killed).

```
renjianxin@raspi00:~/public_html/temp $ mpirun -np 40 --hostfile
raspi-hosts.list ./sobel_mpi_new cathedral.jpg
```

mpirun noticed that process rank 3 with PID 0 on node raspi00 exited on signal 9 (Killed).

```
renjianxin@raspi00:~/public_html/temp $ mpirun -np 40 --hostfile
raspi-hosts.list ./sobel_mpi_new renjianxin.jpg
```

Filtering of input image on Process 1 of raspi00 start ...
Filtering of input image on Process 20 of raspi10 start ...
Filtering of input image on Process 25 of raspi11 start ...
Filtering of input image on Process 33 of raspi13 start ...
Filtering of input image on Process 4 of raspi01 start ...
Filtering of input image on Process 21 of raspi10 start ...
Filtering of input image on Process 36 of raspi14 start ...
Filtering of input image on Process 24 of raspi11 start ...
Filtering of input image on Process 32 of raspi13 start ...
Filtering of input image on Process 16 of raspi04 start ...
Filtering of input image on Process 5 of raspi01 start ...
Filtering of input image on Process 9 of raspi02 start ...
Filtering of input image on Process 37 of raspi14 start ...
Filtering of input image on Process 17 of raspi04 start ...
Filtering of input image on Process 8 of raspi02 start ...
Filtering of input image on Process 2 of raspi00 start ...
Filtering of input image on Process 22 of raspi10 start ...
Filtering of input image on Process 34 of raspi13 start ...
Filtering of input image on Process 6 of raspi01 start ...
Filtering of input image on Process 3 of raspi00 start ...
Filtering of input image on Process 38 of raspi14 start ...
Filtering of input image on Process 26 of raspi11 start ...
Filtering of input image on Process 35 of raspi13 start ...
Filtering of input image on Process 18 of raspi04 start ...
Filtering of input image on Process 10 of raspi02 start ...
Filtering of input image on Process 28 of raspi12 start ...
Filtering of input image on Process 27 of raspi11 start ...
Filtering of input image on Process 12 of raspi03 start ...
Filtering of input image on Process 19 of raspi04 start ...
Filtering of input image on Process 11 of raspi02 start ...
Filtering of input image on Process 0 of raspi00 start ...
Filtering of input image on Process 29 of raspi12 start ...
Filtering of input image on Process 13 of raspi03 start ...
Filtering of input image on Process 30 of raspi12 start ...
Filtering of input image on Process 14 of raspi03 start ...
Filtering of input image on Process 23 of raspi10 start ...
Filtering of input image on Process 7 of raspi01 start ...
Filtering of input image on Process 39 of raspi14 start ...
Filtering of input image on Process 31 of raspi12 start ...
Filtering of input image on Process 15 of raspi03 start ...
Local Min/Max on Process 4: 0/482.004
Local Min/Max on Process 20: 0/482.004
Local Min/Max on Process 25: 0/482.004
Local Min/Max on Process 6: 0/482.004

Local Min/Max on Process 22: 0/482.004
Local Min/Max on Process 24: 0/482.004
Local Min/Max on Process 21: 0/482.004
Local Min/Max on Process 5: 0/482.004
Local Min/Max on Process 27: 0/482.004
Local Min/Max on Process 1: 0/482.004
Local Min/Max on Process 26: 0/482.004
Local Min/Max on Process 7: 0/482.004
Local Min/Max on Process 3: 0/482.004
Local Min/Max on Process 0: 0/482.004
Local Min/Max on Process 23: 0/482.004
Local Min/Max on Process 2: 0/482.004
Local Min/Max on Process 33: 0/482.004
Local Min/Max on Process 34: 0/482.004
Local Min/Max on Process 29: 0/482.004
Local Min/Max on Process 9: 0/482.004
Local Min/Max on Process 17: 0/482.004
Local Min/Max on Process 10: 0/482.004
Local Min/Max on Process 36: 0/482.004
Local Min/Max on Process 12: 0/482.004
Local Min/Max on Process 30: 0/482.004
Local Min/Max on Process 38: 0/482.004
Local Min/Max on Process 35: 0/482.004
Local Min/Max on Process 18: 0/482.004
Local Min/Max on Process 13: 0/482.004
Local Min/Max on Process 31: 0/482.004
Local Min/Max on Process 32: 0/482.004
Local Min/Max on Process 16: 0/482.004
Local Min/Max on Process 28: 0/482.004
Local Min/Max on Process 11: 0/482.004
Local Min/Max on Process 14: 0/482.004
Local Min/Max on Process 37: 0/482.004
Local Min/Max on Process 19: 0/482.004
Local Min/Max on Process 8: 0/482.004
Local Min/Max on Process 15: 0/482.004
Local Min/Max on Process 39: 0/482.004
Global Min/Max: 0/482.004

Filtering of input image on Process 20 finish.
Filtering of input image on Process 22 finish.
Filtering of input image on Process 27 finish.
Filtering of input image on Process 25 finish.
Filtering of input image on Process 24 finish.
Filtering of input image on Process 21 finish.
Filtering of input image on Process 1 finish.

Filtering of input image on Process 26 finish.
Filtering of input image on Process 23 finish.
Filtering of input image on Process 3 finish.
Filtering of input image on Process 2 finish.
Filtering of input image on Process 0 finish.
The total time for execution is 2.41076s
Saved output to renjianxin_local_mpi_out.jpg
Filtering of input image on Process 6 finish.
Filtering of input image on Process 4 finish.
Filtering of input image on Process 7 finish.
Filtering of input image on Process 5 finish.
Filtering of input image on Process 12 finish.
Filtering of input image on Process 13 finish.
Filtering of input image on Process 14 finish.
Filtering of input image on Process 15 finish.
Filtering of input image on Process 17 finish.
Filtering of input image on Process 18 finish.
Filtering of input image on Process 16 finish.
Filtering of input image on Process 19 finish.
Filtering of input image on Process 38 finish.
Filtering of input image on Process 36 finish.
Filtering of input image on Process 37 finish.
Filtering of input image on Process 39 finish.
Filtering of input image on Process 31 finish.
Filtering of input image on Process 29 finish.
Filtering of input image on Process 33 finish.
Filtering of input image on Process 34 finish.
Filtering of input image on Process 30 finish.
Filtering of input image on Process 35 finish.
Filtering of input image on Process 32 finish.
Filtering of input image on Process 28 finish.
Filtering of input image on Process 9 finish.
Filtering of input image on Process 10 finish.
Filtering of input image on Process 11 finish.
Filtering of input image on Process 8 finish.
renjianxin@raspi00:~/public_html/temp \$ mpirun -np 40 --hostfile
raspi-hosts.list ./sobel_mpi_new renjianxin.jpg
Filtering of input image on Process 8 of raspi02 start ...
Filtering of input image on Process 24 of raspi11 start ...
Filtering of input image on Process 16 of raspi04 start ...
Filtering of input image on Process 32 of raspi13 start ...
Filtering of input image on Process 9 of raspi02 start ...
Filtering of input image on Process 0 of raspi00 start ...
Filtering of input image on Process 1 of raspi00 start ...

Filtering of input image on Process 25 of raspi11 start ...
Filtering of input image on Process 17 of raspi04 start ...
Filtering of input image on Process 5 of raspi01 start ...
Filtering of input image on Process 33 of raspi13 start ...
Filtering of input image on Process 21 of raspi10 start ...
Filtering of input image on Process 20 of raspi10 start ...
Filtering of input image on Process 4 of raspi01 start ...
Filtering of input image on Process 37 of raspi14 start ...
Filtering of input image on Process 36 of raspi14 start ...
Filtering of input image on Process 2 of raspi00 start ...
Filtering of input image on Process 10 of raspi02 start ...
Filtering of input image on Process 26 of raspi11 start ...
Filtering of input image on Process 6 of raspi01 start ...
Filtering of input image on Process 22 of raspi10 start ...
Filtering of input image on Process 34 of raspi13 start ...
Filtering of input image on Process 18 of raspi04 start ...
Filtering of input image on Process 38 of raspi14 start ...
Filtering of input image on Process 3 of raspi00 start ...
Filtering of input image on Process 11 of raspi02 start ...
Filtering of input image on Process 7 of raspi01 start ...
Filtering of input image on Process 27 of raspi11 start ...
Filtering of input image on Process 23 of raspi10 start ...
Filtering of input image on Process 35 of raspi13 start ...
Filtering of input image on Process 19 of raspi04 start ...
Filtering of input image on Process 39 of raspi14 start ...
Filtering of input image on Process 12 of raspi03 start ...
Filtering of input image on Process 28 of raspi12 start ...
Filtering of input image on Process 13 of raspi03 start ...
Filtering of input image on Process 29 of raspi12 start ...
Filtering of input image on Process 15 of raspi03 start ...
Filtering of input image on Process 31 of raspi12 start ...
Filtering of input image on Process 14 of raspi03 start ...
Filtering of input image on Process 30 of raspi12 start ...
Local Min/Max on Process 24: 0/482.004
Local Min/Max on Process 21: 0/482.004
Local Min/Max on Process 20: 0/482.004
Local Min/Max on Process 5: 0/482.004
Local Min/Max on Process 6: 0/482.004
Local Min/Max on Process 25: 0/482.004
Local Min/Max on Process 0: 0/482.004
Local Min/Max on Process 27: 0/482.004
Local Min/Max on Process 26: 0/482.004
Local Min/Max on Process 4: 0/482.004
Local Min/Max on Process 2: 0/482.004

Local Min/Max on Process 22: 0/482.004
Local Min/Max on Process 7: 0/482.004
Local Min/Max on Process 23: 0/482.004
Local Min/Max on Process 3: 0/482.004
Local Min/Max on Process 1: 0/482.004
Local Min/Max on Process 10: 0/482.004
Local Min/Max on Process 11: 0/482.004
Local Min/Max on Process 9: 0/482.004
Local Min/Max on Process 8: 0/482.004
Local Min/Max on Process 32: 0/482.004
Local Min/Max on Process 33: 0/482.004
Local Min/Max on Process 18: 0/482.004
Local Min/Max on Process 36: 0/482.004
Local Min/Max on Process 37: 0/482.004
Local Min/Max on Process 16: 0/482.004
Local Min/Max on Process 13: 0/482.004
Local Min/Max on Process 29: 0/482.004
Local Min/Max on Process 30: 0/482.004
Local Min/Max on Process 15: 0/482.004
Local Min/Max on Process 17: 0/482.004
Local Min/Max on Process 12: 0/482.004
Local Min/Max on Process 39: 0/482.004
Local Min/Max on Process 34: 0/482.004
Local Min/Max on Process 35: 0/482.004
Local Min/Max on Process 31: 0/482.004
Local Min/Max on Process 38: 0/482.004
Local Min/Max on Process 19: 0/482.004
Local Min/Max on Process 28: 0/482.004
Local Min/Max on Process 14: 0/482.004
Global Min/Max: 0/482.004

Filtering of input image on Process 21 finish.
Filtering of input image on Process 24 finish.
Filtering of input image on Process 27 finish.
Filtering of input image on Process 20 finish.
Filtering of input image on Process 26 finish.
Filtering of input image on Process 22 finish.
Filtering of input image on Process 25 finish.
Filtering of input image on Process 23 finish.
Filtering of input image on Process 2 finish.
Filtering of input image on Process 3 finish.
Filtering of input image on Process 1 finish.
Filtering of input image on Process 0 finish.

The total time for execution is 2.42285s

Saved output to renjianxin_local_mpi_out.jpg

Filtering of input image on Process 6 finish.
Filtering of input image on Process 7 finish.
Filtering of input image on Process 5 finish.
Filtering of input image on Process 4 finish.
Filtering of input image on Process 16 finish.
Filtering of input image on Process 18 finish.
Filtering of input image on Process 17 finish.
Filtering of input image on Process 19 finish.
Filtering of input image on Process 15 finish.
Filtering of input image on Process 13 finish.
Filtering of input image on Process 12 finish.
Filtering of input image on Process 14 finish.
Filtering of input image on Process 37 finish.
Filtering of input image on Process 36 finish.
Filtering of input image on Process 39 finish.
Filtering of input image on Process 38 finish.
Filtering of input image on Process 30 finish.
Filtering of input image on Process 29 finish.
Filtering of input image on Process 33 finish.
Filtering of input image on Process 32 finish.
Filtering of input image on Process 31 finish.
Filtering of input image on Process 35 finish.
Filtering of input image on Process 28 finish.
Filtering of input image on Process 34 finish.
Filtering of input image on Process 10 finish.
Filtering of input image on Process 11 finish.
Filtering of input image on Process 9 finish.
Filtering of input image on Process 8 finish.
renjianxin@raspi00:~/public_html/temp \$ mpirun -np 40 --hostfile
raspi-hosts.list ./sobel_mpi renjianxin.jpg
Filtering of input image on Process 1 of raspi00 start ...
Filtering of input image on Process 2 of raspi00 start ...
Filtering of input image on Process 3 of raspi00 start ...
Filtering of input image on Process 4 of raspi01 start ...
Filtering of input image on Process 5 of raspi01 start ...
Filtering of input image on Process 6 of raspi01 start ...
Filtering of input image on Process 7 of raspi01 start ...
Filtering of input image on Process 8 of raspi02 start ...
Filtering of input image on Process 9 of raspi02 start ...
Filtering of input image on Process 10 of raspi02 start ...
Filtering of input image on Process 11 of raspi02 start ...
Filtering of input image on Process 12 of raspi03 start ...
Filtering of input image on Process 13 of raspi03 start ...
Min/Max on Process 2 of raspi00: 0/389.597

Send min & max to MASTER ...
Receive global min & max from MASTER ...
Filtering of input image on Process 14 of raspi03 start ...
Min/Max on Process 3 of raspi00: 0/443.644
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 1 of raspi00: 0/427.13
Min/Max on Process 4 of raspi01: 0/450.391
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Filtering of input image on Process 15 of raspi03 start ...
Filtering of input image on Process 16 of raspi04 start ...
Min/Max on Process 5 of raspi01: 0/414.462
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 6 of raspi01: 0/403.205
Send min & max to MASTER ...
Filtering of input image on Process 17 of raspi04 start ...
Receive global min & max from MASTER ...
Filtering of input image on Process 18 of raspi04 start ...
Min/Max on Process 8 of raspi02: 0/421.882
Send min & max to MASTER ...
Min/Max on Process 7 of raspi01: 0/362.322
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 9 of raspi02: 0/366.463
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Filtering of input image on Process 19 of raspi04 start ...
Min/Max on Process 10 of raspi02: 0/364.335
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Filtering of input image on Process 20 of raspi10 start ...
Filtering of input image on Process 21 of raspi10 start ...
Min/Max on Process 11 of raspi02: 0/361.124
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Filtering of input image on Process 22 of raspi10 start ...
Filtering of input image on Process 23 of raspi10 start ...
Filtering of input image on Process 24 of raspi11 start ...
Filtering of input image on Process 25 of raspi11 start ...

Filtering of input image on Process 26 of raspi11 start ...
Filtering of input image on Process 27 of raspi11 start ...
Filtering of input image on Process 28 of raspi12 start ...
Filtering of input image on Process 29 of raspi12 start ...
Filtering of input image on Process 30 of raspi12 start ...
Min/Max on Process 20 of raspi10: 0/335.339
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Filtering of input image on Process 31 of raspi12 start ...
Filtering of input image on Process 32 of raspi13 start ...
Min/Max on Process 21 of raspi10: 0/371.892
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Filtering of input image on Process 33 of raspi13 start ...
Min/Max on Process 22 of raspi10: 0/302.069
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Filtering of input image on Process 34 of raspi13 start ...
Min/Max on Process 23 of raspi10: 0/293.782
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 24 of raspi11: 0/300.585
Send min & max to MASTER ...
Min/Max on Process 25 of raspi11: 0/341.836
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Receive global min & max from MASTER ...
Filtering of input image on Process 35 of raspi13 start ...
Filtering of input image on Process 36 of raspi14 start ...
Min/Max on Process 26 of raspi11: 0/249.87
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Filtering of input image on Process 0 of raspi00 start ...
Min/Max on Process 12 of raspi03: 0/403.091
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 27 of raspi11: 0/282.385
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Filtering of input image on Process 38 of raspi14 start ...
Filtering of input image on Process 37 of raspi14 start ...
Min/Max on Process 13 of raspi03: 0/385.423
Send min & max to MASTER ...
Filtering of input image on Process 39 of raspi14 start ...

Receive global min & max from MASTER ...
Min/Max on Process 14 of raspi03: 0/407.355
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 16 of raspi04: 0/446
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 17 of raspi04: 0/454.348
Send min & max to MASTER ...
Min/Max on Process 15 of raspi03: 0/449.308
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 18 of raspi04: 0/429.677
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 19 of raspi04: 0/429.631
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 0 of raspi00: 0/482.004
Receive min & max from other nodes ...
Min/Max on Process 29 of raspi12: 0/269.706
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 28 of raspi12: 0/331.255
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 30 of raspi12: 0/262.467
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 32 of raspi13: 0/252.153
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 31 of raspi12: 0/263.743
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 33 of raspi13: 0/246.075
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 34 of raspi13: 0/271.78
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 36 of raspi14: 0/316.017
Send min & max to MASTER ...

Receive global min & max from MASTER ...
Min/Max on Process 35 of raspi13: 0/294.957
Send min & max to MASTER ...
Min/Max on Process 39 of raspi14: 0/306.161
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 38 of raspi14: 0/332.936
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 37 of raspi14: 0/259.293
Send min & max to MASTER ...
the global minimum value: 0
the global maximum value: 482.004
Send global min & max to other nodes ...
Receive global min & max from MASTER ...
Filtering of input image on Process 4 of raspi01 finish.
Filtering of input image on Process 2 of raspi00 finish.
Filtering of input image on Process 5 of raspi01 finish.
Filtering of input image on Process 6 of raspi01 finish.
Filtering of input image on Process 1 of raspi00 finish.
Filtering of input image on Process 7 of raspi01 finish.
Filtering of input image on Process 3 of raspi00 finish.
Filtering of input image on Process 10 of raspi02 finish.
Filtering of input image on Process 9 of raspi02 finish.
Filtering of input image on Process 8 of raspi02 finish.
Filtering of input image on Process 22 of raspi10 finish.
Filtering of input image on Process 26 of raspi11 finish.
Filtering of input image on Process 25 of raspi11 finish.
Filtering of input image on Process 27 of raspi11 finish.
Filtering of input image on Process 11 of raspi02 finish.
Filtering of input image on Process 21 of raspi10 finish.
Filtering of input image on Process 20 of raspi10 finish.
Filtering of input image on Process 23 of raspi10 finish.
Filtering of input image on Process 24 of raspi11 finish.
Filtering of input image on Process 0 of raspi00 finish.
Filtering of input image on Process 39 of raspi14 finish.
Filtering of input image on Process 12 of raspi03 finish.
Filtering of input image on Process 29 of raspi12 finish.
Filtering of input image on Process 30 of raspi12 finish.
Filtering of input image on Process 16 of raspi04 finish.
Filtering of input image on Process 32 of raspi13 finish.
Filtering of input image on Process 33 of raspi13 finish.
Filtering of input image on Process 18 of raspi04 finish.

Filtering of input image on Process 14 of raspi03 finish.
Filtering of input image on Process 17 of raspi04 finish.
Filtering of input image on Process 13 of raspi03 finish.
Filtering of input image on Process 19 of raspi04 finish.
Filtering of input image on Process 34 of raspi13 finish.
Filtering of input image on Process 35 of raspi13 finish.
Filtering of input image on Process 15 of raspi03 finish.
Filtering of input image on Process 31 of raspi12 finish.
Filtering of input image on Process 28 of raspi12 finish.
Filtering of input image on Process 36 of raspi14 finish.
Filtering of input image on Process 38 of raspi14 finish.
Filtering of input image on Process 37 of raspi14 finish.
The total time for execution is 0.512237s
renjianxin@raspi00:~/public_html/temp \$ mpirun -np 40 --hostfile
raspi-hosts.list ./sobel_mpi renjianxin.jpg
Filtering of input image on Process 1 of raspi00 start ...
Filtering of input image on Process 2 of raspi00 start ...
Filtering of input image on Process 3 of raspi00 start ...
Filtering of input image on Process 4 of raspi01 start ...
Filtering of input image on Process 5 of raspi01 start ...
Filtering of input image on Process 6 of raspi01 start ...
Filtering of input image on Process 7 of raspi01 start ...
Filtering of input image on Process 8 of raspi02 start ...
Filtering of input image on Process 9 of raspi02 start ...
Filtering of input image on Process 10 of raspi02 start ...
Filtering of input image on Process 11 of raspi02 start ...
Filtering of input image on Process 12 of raspi03 start ...
Filtering of input image on Process 13 of raspi03 start ...
Min/Max on Process 2 of raspi00: 0/389.597
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Filtering of input image on Process 14 of raspi03 start ...
Min/Max on Process 4 of raspi01: 0/450.391
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Filtering of input image on Process 15 of raspi03 start ...
Min/Max on Process 3 of raspi00: 0/443.644
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Filtering of input image on Process 16 of raspi04 start ...
Min/Max on Process 5 of raspi01: 0/414.462
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 6 of raspi01: 0/403.205

Send min & max to MASTER ...
Receive global min & max from MASTER ...
Filtering of input image on Process 17 of raspi04 start ...
Min/Max on Process 7 of raspi01: 0/362.322
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Filtering of input image on Process 18 of raspi04 start ...
Min/Max on Process 1 of raspi00: 0/427.13
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Filtering of input image on Process 19 of raspi04 start ...
Filtering of input image on Process 20 of raspi10 start ...
Filtering of input image on Process 21 of raspi10 start ...
Filtering of input image on Process 22 of raspi10 start ...
Filtering of input image on Process 23 of raspi10 start ...
Filtering of input image on Process 24 of raspi11 start ...
Filtering of input image on Process 25 of raspi11 start ...
Filtering of input image on Process 26 of raspi11 start ...
Filtering of input image on Process 27 of raspi11 start ...
Filtering of input image on Process 28 of raspi12 start ...
Filtering of input image on Process 29 of raspi12 start ...
Filtering of input image on Process 30 of raspi12 start ...
Filtering of input image on Process 31 of raspi12 start ...
Filtering of input image on Process 32 of raspi13 start ...
Min/Max on Process 20 of raspi10: 0/335.339
Send min & max to MASTER ...
Min/Max on Process 21 of raspi10: 0/371.892
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Receive global min & max from MASTER ...
Filtering of input image on Process 33 of raspi13 start ...
Min/Max on Process 8 of raspi02: 0/421.882
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 22 of raspi10: 0/302.069
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Filtering of input image on Process 34 of raspi13 start ...
Min/Max on Process 9 of raspi02: 0/366.463
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 24 of raspi11: 0/300.585
Send min & max to MASTER ...
Receive global min & max from MASTER ...

Min/Max on Process 23 of raspi10: 0/293.782
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 10 of raspi02: 0/364.335
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Filtering of input image on Process 35 of raspi13 start ...
Filtering of input image on Process 36 of raspi14 start ...
Filtering of input image on Process 0 of raspi00 start ...
Min/Max on Process 25 of raspi11: 0/341.836
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 26 of raspi11: 0/249.87
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Filtering of input image on Process 37 of raspi14 start ...
Filtering of input image on Process 38 of raspi14 start ...
Filtering of input image on Process 39 of raspi14 start ...
Min/Max on Process 27 of raspi11: 0/282.385
Send min & max to MASTER ...
Min/Max on Process 12 of raspi03: 0/403.091
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 11 of raspi02: 0/361.124
Send min & max to MASTER ...
Min/Max on Process 13 of raspi03: 0/385.423
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 14 of raspi03: 0/407.355
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 16 of raspi04: 0/446
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 15 of raspi03: 0/449.308
Send min & max to MASTER ...
Min/Max on Process 17 of raspi04: 0/454.348
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 18 of raspi04: 0/429.677
Send min & max to MASTER ...

Receive global min & max from MASTER ...
Min/Max on Process 0 of raspi00: 0/482.004
Receive min & max from other nodes ...
Min/Max on Process 19 of raspi04: 0/429.631
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 28 of raspi12: 0/331.255
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 29 of raspi12: 0/269.706
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 31 of raspi12: 0/263.743
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 30 of raspi12: 0/262.467
Send min & max to MASTER ...
Min/Max on Process 33 of raspi13: 0/246.075
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 32 of raspi13: 0/252.153
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 34 of raspi13: 0/271.78
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 36 of raspi14: 0/316.017
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 35 of raspi13: 0/294.957
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 37 of raspi14: 0/259.293
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 38 of raspi14: 0/332.936
Send min & max to MASTER ...
Receive global min & max from MASTER ...
Min/Max on Process 39 of raspi14: 0/306.161
Send min & max to MASTER ...
the global minimum value: 0
the global maximum value: 482.004
Send global min & max to other nodes ...

Receive global min & max from MASTER ...
Filtering of input image on Process 0 of raspi00 finish.
Filtering of input image on Process 2 of raspi00 finish.
Filtering of input image on Process 4 of raspi01 finish.
Filtering of input image on Process 3 of raspi00 finish.
Filtering of input image on Process 5 of raspi01 finish.
Filtering of input image on Process 26 of raspi11 finish.
Filtering of input image on Process 6 of raspi01 finish.
Filtering of input image on Process 1 of raspi00 finish.
Filtering of input image on Process 22 of raspi10 finish.
Filtering of input image on Process 21 of raspi10 finish.
Filtering of input image on Process 23 of raspi10 finish.
Filtering of input image on Process 27 of raspi11 finish.
Filtering of input image on Process 7 of raspi01 finish.
Filtering of input image on Process 25 of raspi11 finish.
Filtering of input image on Process 24 of raspi11 finish.
Filtering of input image on Process 20 of raspi10 finish.
Filtering of input image on Process 39 of raspi14 finish.
Filtering of input image on Process 29 of raspi12 finish.
Filtering of input image on Process 10 of raspi02 finish.
Filtering of input image on Process 33 of raspi13 finish.
Filtering of input image on Process 8 of raspi02 finish.
Filtering of input image on Process 34 of raspi13 finish.
Filtering of input image on Process 12 of raspi03 finish.
Filtering of input image on Process 13 of raspi03 finish.
Filtering of input image on Process 18 of raspi04 finish.
Filtering of input image on Process 15 of raspi03 finish.
Filtering of input image on Process 9 of raspi02 finish.
Filtering of input image on Process 17 of raspi04 finish.
Filtering of input image on Process 16 of raspi04 finish.
Filtering of input image on Process 14 of raspi03 finish.
Filtering of input image on Process 19 of raspi04 finish.
Filtering of input image on Process 37 of raspi14 finish.
Filtering of input image on Process 36 of raspi14 finish.
Filtering of input image on Process 38 of raspi14 finish.
Filtering of input image on Process 11 of raspi02 finish.
Filtering of input image on Process 28 of raspi12 finish.
Filtering of input image on Process 31 of raspi12 finish.
Filtering of input image on Process 30 of raspi12 finish.
Filtering of input image on Process 32 of raspi13 finish.
Filtering of input image on Process 35 of raspi13 finish.
The total time for execution is 0.368396s